

PANFENG JIANG

(332) 201-2479 | pj2479@columbia.edu | jiangpf2022.github.io | linkedin.com/in/pfjiang

EDUCATION

Columbia University

M.S. in Computer Science

Courses: Databases, LLM-Based Generative AI

New York, NY

Aug 2026 – Expected Dec 2027

ShanghaiTech University

B.E. in Computer Science, GPA: 3.75/4.0, Rank: 11/161

Recipient of National Scholarship, Outstanding Graduate of Shanghai, ShanghaiTech Outstanding Student Award

Courses: Probability & Statistics, Algorithms, Linear Algebra, Computer Vision, Machine Learning

Shanghai, China

Sep 2022 – Jun 2026

University of California, Berkeley

Undergraduate Exchange Program in Computer Science, GPA: 3.8/4.0

Recipient of 2025 ShanghaiTech International Exchange Program Scholarship

Courses: Optimization, Deep Learning, Artificial Intelligence, LLM Agents, Abstract Algebra

Berkeley, CA

Aug 2024 – Jun 2025

PROJECT EXPERIENCE

Bay Area Data Science Summit

Project: MLP for Optimal Equity Derivative Hedging

- Led the team to win 1st Place in the UC Berkeley competition endorsed by Wells Fargo.
- Collaborated with teammates from 3 countries to build an MLP-based pipeline for equity derivative hedging.
- Cut test-set premium cost by 64.5% versus baseline, while maintaining \$10M exposure target.

Berkeley, CA

Feb 2025 – Mar 2025

Mathematical Contest in Modeling

Project: Rebalancing Nature's Scale, A Model to Tame Overtourism

- Earned Outstanding Winner Award and American Mathematical Society Award, Top 3 among 21,054 teams.
- Proposed a coupled ODE model integrating 3 dimensions: economic, environmental, and social dynamics.
- Optimized intervention and tax allocation policies through simulation and AHP evaluation.

Berkeley, CA

Feb 2025 – Mar 2025

EECS 182: Deep Learning, UC Berkeley

Project: Beyond the LRH in LLMs, Controlled Perturbations and Concept Subspaces

- Identified limitations of widely used LRH and proposed multidimensional representations for complex semantics.
- Designed a new concept-subspace steering approach beyond single-direction representation.
- Increased concept alignment from 0.5/5 to 5.0/5 by expanding latent steering from one to five directions.

Berkeley, CA

Mar 2025 – May 2025

PROFESSIONAL EXPERIENCE

Columbia University, Creative Machines Lab

Research Assistant

- Studying whether LLM confidence and correctness can be inferred from internal neural activations.
- Training supervisory models to detect unreliable or deceptive LLM responses from hidden representations.

New York, NY

Sep 2026 – Present

UNC Chapel Hill, Weitong Zhang's Lab

Research Assistant

- Coauthored a manuscript on variance-aware Langevin Monte Carlo Thompson Sampling.
- Spearheaded an LMC Thompson Sampling extension for bandits with action-dependent noise variance.
- Derived regret bounds quantifying how finite-step posterior error affects exploration efficiency.

Remote

Apr 2025 – Oct 2025

Roboscience Co., Ltd. – NUS Lin Shao's Lab Collaboration

Artificial Intelligence Intern

- Led research resulting in a co-first-author paper accepted to IEEE ICRA 2026.
- Created a VLM-based navigation framework combining high-level reasoning with real-time motion control.
- Achieved 91.3% navigation success versus 28% for prior state-of-the-art method.

Shenzhen, China

Jun 2025 – Sep 2025

ShanghaiTech University, Mobile Perception Lab

Research Assistant

- Coauthored an ICCV 2025 Highlight paper on asynchronous structure and motion.
- Developed asynchronous detection and point-tracking modules for event-based motion estimation.
- Achieved 63 μ s CPU runtime and 10–30% lower velocity error with an event-based solver.

Shanghai, China

Apr 2023 – Mar 2025

TECHNICAL SKILLS

Programming Languages: Python, C++, MATLAB, SQL

ML & Scientific Computing: PyTorch, NumPy, Pandas, Scikit-learn, Matplotlib

Development Tools: Linux, Git, Jupyter, L^AT_EX, ROS

Technical Areas: Machine Learning, Optimization, Statistical Modeling, Numerical Computing